AIRLINE_DATASET_DB PROJECT TASK

1. Create a Database in PostgreSQL named "Airline_Dataset_db"
2. Create an appropriate table structure named "airline_data_tb" and import the entire data from the "Airline Dataset Updated v2.csv" file.
3. You are to provide the management of the airline data analysis of the dataset and also provide some visual representation, write at least 5 SQL commands/statements to extract the required information.

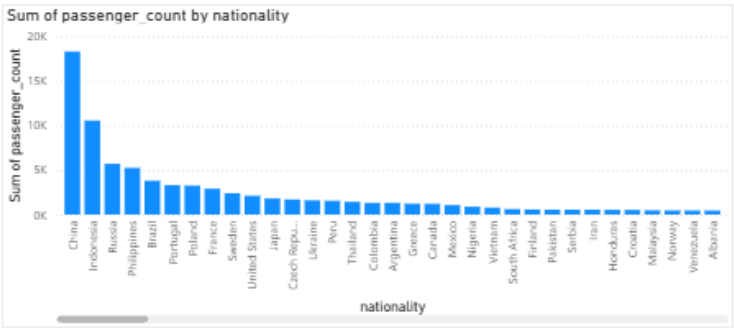
```
DROP TABLE IF EXISTS airline_data_tb;
-- 2 Creating the table structure --
CREATE TABLE airline_data_tb (
   Passenger_ID VARCHAR(50) PRIMARY KEY,
   First_Name VARCHAR(50),
   Last_Name VARCHAR(50),
   Gender VARCHAR(10),
   Age INT,
   Nationality VARCHAR(50),
   Airport_Name VARCHAR(100),
   Airport_Country_Code VARCHAR(10),
   Country_Name VARCHAR(50),
   Airport_Continent VARCHAR(20),
   Continents VARCHAR(50),
   Departure_Date DATE,
   Arrival_Airport VARCHAR(100),
   Pilot_Name VARCHAR(50),
   Flight_Status VARCHAR(20)
);


-- Set datestyle before import to avoid error
SET datestyle = 'MDY';

-- Importing the Airline dataset --
COPY airline_data_tb(
   Passenger_ID, First_Name, Last_Name, Gender, Age, Nationality,
   Airport_Name, Airport_Country_Code, Country_Name, Airport_Continent,
   Continents, Departure_Date, Arrival_Airport, Pilot_Name, Flight_Status
)
FROM 'C:/ProgramData/MySQL/MySQL Server
8.0/Uploads/input_files/Airline_Dataset_v2.csv'
DELIMITER ','
CSV HEADER;
```

-- See all the records loaded
SELECT * FROM airline_data_tb
LIMIT 20;

f age	airport_continent	airport_country_code	airport_name	arrival_airport	continents	country_name	Year	Quarter	Month	Day	first_name	flight_status
22	AF	ZA	Oudtshoorn Airport	OUH	Africa	South Africa	2022	Qtr 3	July	18	Dyna	Cancelled
12	AS	CN	Enshi Airport	ENH	Asia	China	2022	Qtr 1	March	29	Alvin	Delayed
62	AS	CN	Guilin Liangjiang International Airport	KWL	Asia	China	2022	Qtr 2	May	9	Warner	Cancelled
47	AS	PK	Loralai Airport	LRG	Asia	Pakistan	2022	Qtr 1	March	19	Kalie	Delayed
67	EU	FR	Grenoble-Isère Airport	GNB	Europe	France	2022	Qtr 1	January	18	Darby	On Time
73	EU	GB	Duxford Aerodrome	QFO	Europe	United Kingdom	2022	Qtr 4	October	30	Rand	Cancelled
35	EU	IT	Venice Marco Polo Airport	VCE	Europe	Italy	2022	Qtr 3	August	20	Aleda	On Time
62	NAM	CA	Kugluktuk Airport	YCO	North America	Canada	2022	Qtr 4	December	26	Elwood	On Time
71	NAM	CA	Ottawa / Gatineau Airport	YND	North America	Canada	2022	Qtr 3	September	16	Dominica	Delayed
87	NAM	CA	Thompson Airport	YTH	North America	Canada	2022	Qtr 1	March	11	Jerrine	Cancelled
13	NAM	CA	Vermilion Airport	YVG	North America	Canada	2022	Qtr 2	June	4	Burlie	On Time
39	NAM	MX	Nuevo Casas Grandes Airport	NCG	North America	Mexico	2022	Qtr 2	May	27	Porty	Cancelled
71	NAM	PA	Ruben Cantu Airport	SYP	North America	Panama	2022	Qtr 2	June	2	Briant	Delayed
62	NAM	US	Coldfoot Airport	CXF	North America	United States	2022	Qtr 2	June	28	Edithe	On Time
32	NAM	US	Farmington Regional Airport	FAM	North America	United States	2022	Qtr 3	July	3	Amberly	Delayed
21	NAM	US	Gillespie Field	SEE	North America	United States	2022	Qtr 1	February	25	Bay	On Time
77	OC	AU	Cudal Airport	CUG	Oceania	Australia	2022	Qtr 1	March	24	Catriona	Delayed
55	SAM	BR	Coronel Horácio de Mattos Airport	LEC	South America	Brazil	2022	Qtr 4	October	6	Lora	On Time
36	SAM	BR	Maestro Wilson Fonseca Airport	STM	South America	Brazil	2022	Qtr 3	July	4	Perceval	Cancelled
28	SAM	VE	Zaraza Airport	ZRZ	South America	Venezuela, Bolivarian Republic of	2022	Qtr 3	September	23	Janella	On Time
972												

-- Count the total number of records loaded
SELECT COUNT (*)
FROM airline_data_tb;

-- 3. Here are 5 useful queries for airline management analysis:
-- 1 Passenger count by nationality
SELECT nationality, COUNT(*) AS passenger_count
FROM airline_data_tb
GROUP BY nationality
ORDER BY passenger_count DESC;



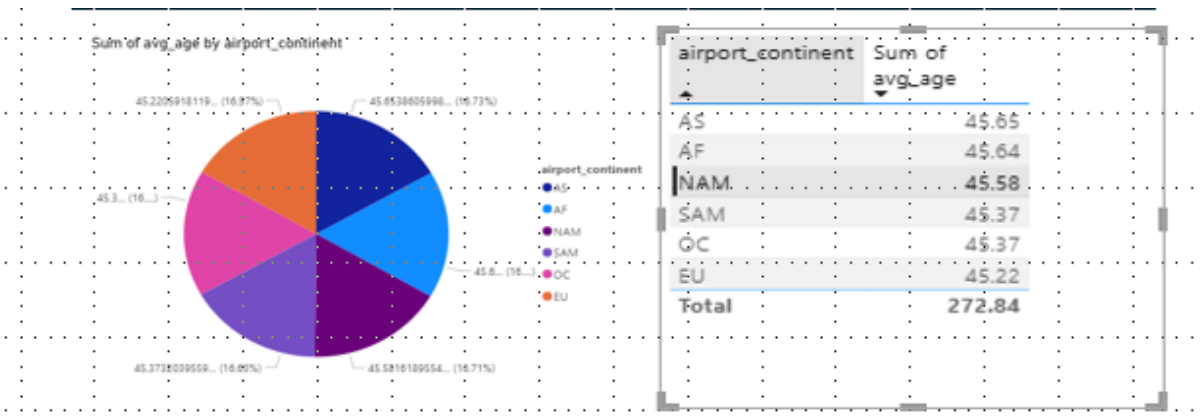
nationality	Sum of passenger_count
China	18317
Indonesia	10559
Russia	5693
Philippines	5239
Brazil	3791
Portugal	3299
Poland	3245
France	2907
Sweden	2397
United States	2105
Japan	1805
Czech Republic	1690
Ukraine	1591
Peru	1540
Thailand	1426
Colombia	1310
Argentina	1308
Greece	1213
Canada	1198
Mexico	1054
Nigeria	880
Vietnam	776
Total	**98619**

Sum of passenger_count was highest for China at 18317, followed by Indonesia and Russia.

China accounted for 18.57% of Sum of passenger_count.

Across all 240 nationality, Sum of passenger_count ranged from 1 to 18317.

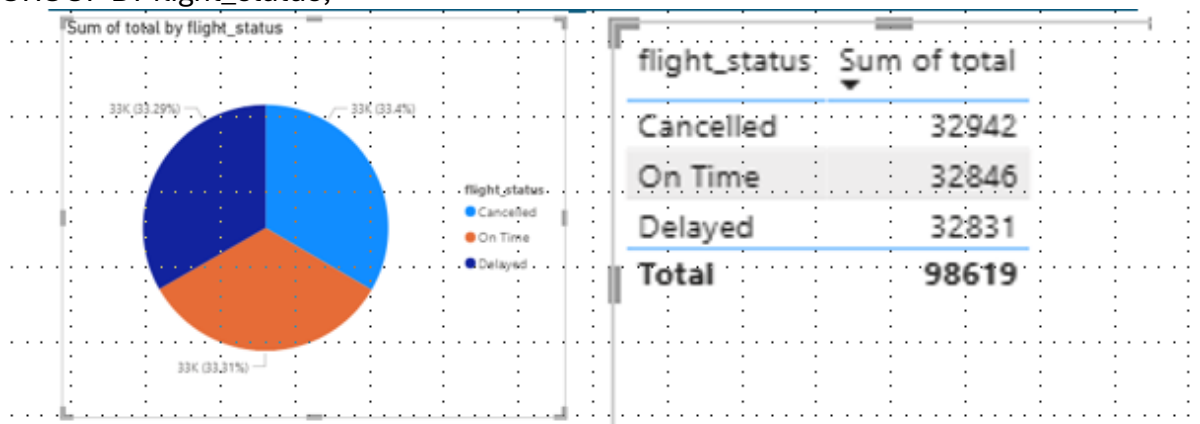
-- 2 Average age of passengers per continent
SELECT airport_continent, AVG(age) AS avg_age
FROM airline_data_tb;



North America (NAM): 45.58 years, Africa (AF): 45.63 years, Asia (AS): 45.65 years, Oceania (OC): 45.37 years
South America (SAM): 45.37 years, Europe (EU): 45.22 years.
Asia has the highest average (45.65), while Europe has the lowest (45.22).

GROUP BY airport_continent;

-- 3 Flight status distribution
SELECT flight_status, COUNT(*) AS total
FROM airline_data_tb
GROUP BY flight_status;



Cancelled had the highest Sum of total at 32942, followed by On Time at 32846 and Delayed at 32831. Cancelled accounted for 33.40% of Sum of total.

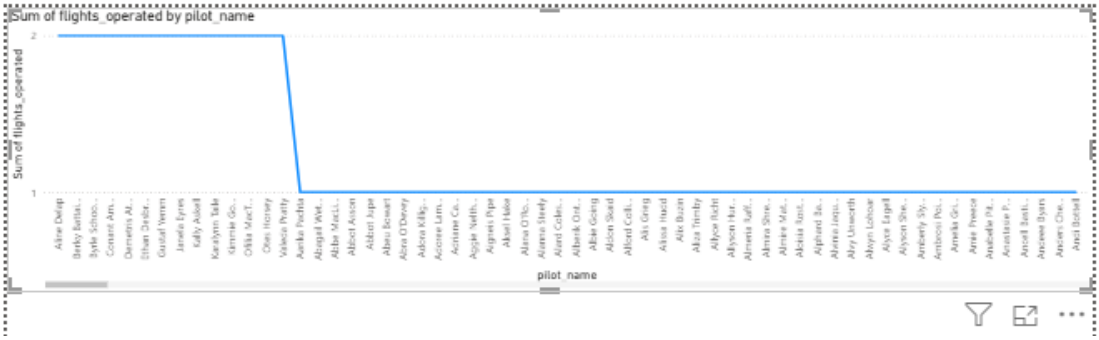
-- 4. Top 5 busiest airports (by departures)
SELECT airport_name, COUNT(*) AS departures
FROM airline_data_tb
GROUP BY airport_name
ORDER BY departures DESC
LIMIT 5;

airport_name	Sum of departures
Böblingen Flugfeld	36
San Fernando Airport	31
San Pedro Airport	43
Santa Ana Airport	35
Santa Maria Airport	38
Total	**183**

At 43, San Pedro Airport had the highest Sum of departures and was 38.71% higher than San Fernando Airport, which had the lowest Sum of departures at 31. San Pedro Airport accounted for 23.50% of Sum of departures. Across all 5 airport name, Sum of departures ranged from 31 to 43.

-- 5 Number of flights handled by each pilot

```
SELECT pilot_name, COUNT(*) AS flights_operated
FROM airline_data_tb
GROUP BY pilot_name
ORDER BY flights_operated DESC;
```




```
-- Flights per departure date
SELECT Departure_Date, COUNT(*) AS flights_count
FROM airline_data_tb
GROUP BY Departure_Date
ORDER BY Departure_Date;
```

Year	Quarter	Month	Day	Sum of flights_count
2022	Qtr 1	January	1	267
2022	Qtr 1	January	2	283
2022	Qtr 1	January	3	262
2022	Qtr 1	January	4	263
2022	Qtr 1	January	5	302
2022	Qtr 1	January	6	236
2022	Qtr 1	January	7	272
2022	Qtr 1	January	8	292
2022	Qtr 1	January	9	275
2022	Qtr 1	January	10	269
2022	Qtr 1	January	11	268
2022	Qtr 1	January	12	253
2022	Qtr 1	January	13	301
2022	Qtr 1	January	14	280
2022	Qtr 1	January	15	263
2022	Qtr 1	January	16	301
2022	Qtr 1	January	17	293
2022	Qtr 1	January	18	276
2022	Qtr 1	January	19	269
2022	Qtr 1	January	20	277
2022	Qtr 1	January	21	282
2022	Qtr 1	January	22	266
2022	Qtr 1	January	23	281
2022	Qtr 1	January	24	261
2022	Qtr 1	January	25	255
2022	Qtr 1	January	26	280
2022	Qtr 1	January	27	268
2022	Qtr 1	January	28	304
2022	Qtr 1	January	29	242
2022	Qtr 1	January	30	257
2022	Qtr 1	January	31	261
2022	Qtr 1	February	1	249
2022	Qtr 1	February	2	286
Total				**98619**

COMMIT;

-- VISUAL REPRESENTATIONS
-- I visualized these queries using tools like Power BI

-- Bar chart: Passenger count by nationality
-- Pie Chart: Average age of passengers per continent
-- Pie chart: Flight status distribution
-- Bar chart: Top 5 busiest airports (by departures)
-- Line Chart: Number of flights handled by each pilot
-- Line chart: Flights per departure date